

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 15

(1) Research Program/Project/Study Title: Characterization of Paddy Soils in Capiz: Basis for Soil Quality Improvement Program	
Proponent Name:	Dr. Ryan T. Sarimong
Academic Rank:	Professor IV
Co-Authors (with academic rank):	
Campus/Satellite College/Address:	Burias Campus
Telephone/Cell Phone/Fax Number:	09977847068
E-mail Address:	rtsarimong@capsu.edu.ph

Program or Project has component (if no proceed to **2**)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/BiologicalSciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
--	--

(4) Site of implementation			
		Municipality/City:	Mambusao
		Province:	Capiz
		Region:	Western Visayas
(5) Classification		(6) Mode of Implementation	
Research:	Development:	☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
☐ Basic	☐ Pilot Testing		
☒ Applied	☐ Tech. Promotion/Commercialization		

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 2 of 15

(7) Sector/Commodity	(8) Discipline/Field of Specialization
Agriculture	Crop, Soil and Environmental Sciences

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Soil survey and characterization was conducted in four identified paddy sites to provide a detailed soil profile description and landscape assessment of the selected paddy soils, determine the physical and chemical properties of selected paddy soils; and come up with a soil quality improvement program for paddy soils in Capiz. Soil samples were collected and processed. Soil physical properties which were determined included soil texture, soil structure, soil color, bulk density and soil porosity. The chemical properties of soil analyzed were pH, soil organic matter, total nitrogen (N), available phosphorus (P), exchangeable potassium (K), and electrical conductivity (EC). The physical and chemical properties of top soil and subsoil of the four sites were interpreted using their means and compared with the established values for different soil parameters indicative of a paddy soil. Results revealed that the different paddy soils under study varied in terms of their horization and morphological features. Paddy soils which were characterized in this study differed in their physical and chemical properties attributed to the type of cultural and management practices employed by the tenant farming the land. Thus, it is imperative to employ a package of management recommendations to address the limitations and properties identified in each site and to make the soil more favorable for rice production.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

paddy, soil horizons, soil profile, soil properties, soil quality

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Soil quality is the ability of a soil to perform functions that are essential to people and the environment. In simple terms, soil quality is how well soil does what we want it to do. More specifically, soil quality is the capacity of a specific kind of soil to function, within natural or managed ecosystem boundaries, to sustain plant and animal productivity, maintain or enhance water and air quality, and support human health and habitation (USDA NRCS, und.).

Soil has both inherent and dynamic qualities. Inherent soil quality is a soil's natural ability to function. These characteristics do not change easily. Dynamic soil quality is how soil changes depending on how it is managed.

One goal of soil quality research is to learn how to manage soil in a way that improves soil function. Soils respond differently to management depending on the inherent properties of the soil and the surrounding landscape.

Yet, it is fundamental to first assess the soil physical, chemical and biological properties to know its most appropriate suitability for a specific purpose. Soil characterization is also an important component in land use classification. When the soil is properly analyzed and identified, the risks of incompatible management recommendations will be lessened and selection of knowledge and technologies to apply will be efficient. Hence, this study was undertaken.

Generally, this study aimed to characterize some paddy soils in Capiz. Specifically, it attempted to (1) provide a detailed soil profile description and landscape assessment of the selected paddy soils in

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 3 of 15

Capiz; (2) determine the physical and chemical properties of selected paddy soils in Capiz; and (3) come up with a soil quality improvement program for paddy soils in Capiz.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Site Selection for Soil Characterization

Based on the soil map of Capiz, the various soil series were noted and on-site validation through reconnaissance soil survey was done to know the current land use of each soil series. The location and coordinate of the candidate paddy sites were determined and noted through GPS application.

Courtesy Call and Securing of Permits

Once the sampling sites were identified, courtesy calls were made to the respective barangay captains. In order to dig pits and collect soil samples for characterization, permission was obtained from the landowners or tenants.

Site Visit and Interview

Once permission had been obtained from the landowner or tenant, the area was visited to observe any potential physical and chemical characteristics, and assess the condition of the land. Interview with the landowner or tenant was also conducted regarding the farming history and the status of the land.

Soil Sampling and Profile Description

Soil profiling and characterization process involved identifying and visiting the designated area. A pit was excavated, with dimensions of 1 meter in length, 1 meter in width, and a depth of 1.5 meters or more, depending on the site conditions. Upon digging the pit, the soil profile was examined by carefully scraping the soil layers with a knife, starting from the top. This helped determine the variations and properties between different soil horizons. Sticks were placed to mark the different horizons where soil properties and colors changed. The depths of each horizon were measured using a tape measure. For each sampling site, a detailed soil profile characterization and landscape description were done. Composite soil samples were also collected for analysis.

Soil Sample Preparation

Soil samples were collected from each identified soil horizon. For the chemical characteristics analysis, three samples were taken from each horizon, with each sample consisting of one kilogram of soil cut with a knife. Additionally, three separate samples were taken from each horizon using a can to determine physical characteristics such as bulk density and porosity.

To preserve the samples for chemical analysis, they were placed in labeled plastic bags and transported to the laboratory for air-drying over a period of three days. After air drying, the samples were pulverized and sifted through a sieve with 2mm diameter holes. The soil material that passed through the sieve was thoroughly mixed and stored in a plastic container. This composite sample served as the stock sample submitted for soil analysis.

For the physical characteristics analysis, cans containing the soil sample were transported to the laboratory. Soil samples were placed in an oven and underwent drying for 24 hours to determine bulk density and porosity. The diameter and height of the can used in sampling was likewise determined and used in the computation of the volume of soil.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 4 of 15

Data Collection and Interpretation

The following data were gathered:

1. Soil profile characterization – horizon, depth, color, mottles, texture, structure, consistence, plasticity, roots, pores, concretions, rock fragments, pH, soil macrofauna and boundary
2. Landscape description – location of pedon, physiographic position, elevation, soil moisture regime, soil temperature regime, present vegetation, evidence of erosion, parent material, surrounding areas, drainage, moisture, rock class and human influence
3. Physical properties – particle size distribution using hydrometer method, bulk density using core method, percent porosity
4. Chemical properties – pH using potentiometer, organic matter using Walkley-Black method, total N using modified Kjeldahl method, available P using Troug method, exchangeable K, Ca and Mg using MA-AES method, electrical conductivity using EC Meter 2510 B

These data were analyzed and interpreted using the soil fertility criteria of PCAARRD (2013). The results of interpretation served as basis for recommendation of management strategies in the development of soil quality improvement program.

(12) **Results and Discussions** (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

SITE 1

General Description of Study Site 1

Site 1 was located at Sitio Aliwas, Burias, Mambusao, Capiz with coordinates 11⁰ 26' 53" N latitude, 122⁰ 32' 54" E longitude. It was surrounded by rice field, river, houses and vegetable production area (Figure 1.0). There was no evidence of erosion in the area because it has a flat topography.



Figure 1.0. Study site 1.

An interview with conventional farmers revealed that all of them had been cultivating rice for three to four years. All of them used fertilizers on their rice plants. The inorganic fertilizers they applied were complete fertilizer (14-14-14), urea (46-0-0) and ammonium phosphate (16-20-0). All of the farmers applied pesticides on their rice plants. Brodan, lanette, nativo and magnum were the pesticides they used. The average yield per cropping ranged from 25-30 sacks. Furthermore, farm activities were carried out by humans, animals, and machinery.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 5 of 15

Site 1 belongs to Alimodian clay soil type. Soil profile information and soil profile description of Site 1 are presented in Table 1.0 and Table 2.0, respectively. According to Carating et al. (2014), Alimodian soil series is classified as Typic Hapludalfs. The soil of this series is derived from the weathered products of shale and sandstone, with shale dominating. The surface soil is brown to reddish brown clay loam, good medium granular structure, slightly friable when moist and brittle when dry. The subsoil is light brown clay loam, weak medium columnar structure, slightly brittle and slightly compact.

Table 1.0. Soil profile information of Site 1.

Profile Number	1
Soil Name	Conventional paddy
Soil Classification	Typic Hapludalfs*
Location of Pedon	Sitio Aliwas, Burias, Mambusao, Capiz (11° 26' 53" N, 122° 32' 54" E)
Physiographic Position	Toeslope
Elevation	33 meters above sea level
Soil Moisture Regime	Udic
Soil Temperature Regime	Isohyperthermic
Present Vegetation	Rice
Farmland	Rice field
Evidence of Erosion	None
Parent Material	Shale
Surrounding Areas	Rice field, river, houses, vegetable production area
Drainage	Poor
Moisture (depths)	Moist from the surface up to 20 cm; presence of aquifer at 100 cm
Stones / Rock class	None
Human Influence	Cultivation and farming
Date	August 7, 2021

*Adapted from Carating et al. (2014)

Table 2.0. Soil profile description of Site 1.

Horizon Designation	Depth	Horizon Description
Aptg	0-26 cm	Dark brown (10 YR 3/3) moist; clay; common red mottles; strong, coarse granular structure; firm, sticky; many fine roots; few micropores to fine irregular pores; strongly acidic (pH 5.01); medium nitrogen; very low phosphorus; moderately unfavorable potassium; clear smooth boundary.
Btg1	26-38 cm	Brown (10 YR 3/4) moist; clay; common red mottles; moderate, coarse prismatic structure; firm, slightly sticky; common very fine roots; few micropores to very fine irregular pores; slightly acidic (pH 6.37); medium nitrogen; medium phosphorus; moderately unfavorable potassium; clear smooth boundary.
Btg2	below 38 cm	Brownish gray (10 YR 4/1) moist; clay; few red mottles; moderate, coarse blocky angular structure; firm, slightly sticky; few very fine roots; few micropores to very fine irregular pores; slightly acidic (pH 6.56); medium nitrogen; medium phosphorus; moderately unfavorable potassium; clear smooth boundary.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 6 of 15

Soil Physical and Chemical Properties in Site 1

The physical and chemical properties of Site 1 are presented in Table 3.0. In terms of bulk density, the subsoil is more compact as compared to the topsoil. As to the percent pore space, the top soil is more porous than the subsoil. These findings could be attributed to the farmers practice like the frequent use of machines in farm operations which possibly led to the compaction of soil.

In terms of chemical properties, the topsoil is more acidic than the subsoil. This could be attributed to the applied commercial inorganic fertilizer which contribute to the residual soil acidity. Organic matter and total nitrogen were both higher in the topsoil than the subsoil, while available phosphorus and exchangeable potassium were both higher in the subsoil as compared to the topsoil.

Table 3.0. Physical and chemical properties of soil in Site 1.

Parameters	Values Obtained	
	Top soil	Subsoil
Bulk density (g cm ⁻³)	1.38	1.42
Porosity (%)	48.00	46.50
pH	5.01	6.37
Organic matter (%)	4.37	1.77
Total Nitrogen (%)	0.50	0.33
Available P (ppm)	1.33	22.83
Exchangeable K (ppm)	78.67	109.83

SITE 2

General Description of Study Site 2

The paddy for Site 2 was located at CAPSU, Burias, Mambusao, Capiz, with coordinates 11° 26' 35" N latitude, 122° 32' 54" E longitude. It was surrounded by houses, buildings, rice field, fish pond and vegetable area (Figure 2.0). The drainage in the area was poor for it has a flat topography.



Figure 2.0. Study site 2.

Interview with farmers engaged in organic farming manifested that most of them cultivated rice for more than two years. All of them applied fertilizers to their rice plants. The organic fertilizers they applied were bokashi and vermitea. All of them did not apply synthetic pesticide. Their average yield per cropping was between 12-16 sacks. In addition, most of the farm activities were done by humans, animals and machinery.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 7 of 15

Site 2 belongs to Alimodian clay soil type. Table 4.0 and Table 5.0 presents the soil profile information and soil profile description, respectively of Site 2. According to Carating et al. (2014), Alimodian soil series is classified as Typic Hapludalfs. The Alimodian series was initially described in the municipality of Alimodian in the province of Iloilo. This series' soil is composed of worn shale and sandstone, with shale predominant. The surface soil is a brown to reddish brown clay loam with an excellent medium granular structure that is slightly friable when wet and brittle when dry. The subsoil is a light brown clay loam with a weak medium columnar structure, is slightly brittle, and is slightly compact.

Table 4.0. Soil profile information of Site 2.

Profile Number	2
Soil Name	Organic paddy
Soil Classification	Typic Hapludalfs*
Location of Pedon	CAPSU, Burias, Mambusao, Capiz (11° 26' 35" N, 122° 32' 54" E)
Physiographic Position	Toeslope
Elevation	28 meters above sea level
Soil Moisture Regime	Udic
Soil Temperature Regime	Isohyperthermic
Present Vegetation	Rice
Farmland	Rice field
Evidence of Erosion	None
Parent Material	Shale
Surrounding Areas	Rice field, building, houses, vegetable production, fish pond
Drainage	Poor
Moisture (depths)	Moist from the surface up to 20 cm; presence of aquifer at 92 cm
Stones / Rock class	None
Human Influence	Cultivation, land preparation, planting
Date	August 22, 2021

*Adapted from Carating et al. (2014)

Table 5.0. Soil profile description of Site 2.

Horizon Designation	Depth	Horizon Description
Aptg	0-24 cm	Grayish brown (7.5 YR 4/2) moist; clay; few red mottle; strong, coarse granular structure; very firm, very sticky; many fine roots; few micropores to fine irregular pores; slightly acidic (pH 6.12); low nitrogen; high phosphorus; moderately unfavorable potassium; clear smooth boundary.
Btg1	24- 83cm	Brown (7.5 YR 3/4) moist; clay; few red mottles; moderate, coarse prismatic structure; friable, slightly sticky; few very fine roots; few micropores to very fine irregular pores; slightly acidic (pH 6.54); low nitrogen; low phosphorus; favorable potassium; clear smooth boundary.
Btg2	below 83 cm	Brownish black (7.5 YR 3/2) moist; clay; few red mottles; moderate, coarse blocky angular structure; friable, slightly sticky; few very fine roots; few micropores to very fine irregular pores; near neutral (pH 6.70); low nitrogen; low phosphorus; favorable potassium; clear smooth boundary.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 8 of 15

Soil Physical and Chemical Properties in Site 2

The physical and chemical properties of Site 2 are presented in Table 6.0. In terms of bulk density, the subsoil is more compact as compared to the topsoil. As to the percent pore space, the top soil is more porous than the subsoil. These findings could be attributed to the farmers practice like the frequent use of machines in farm operations which possibly led to the compaction of soil. Also, the application of organic fertilizers contribute to a more porous surface soil due to the formation of stable aggregates.

In terms of chemical properties, the topsoil is more acidic than the subsoil. This could be attributed to the continued decomposition of organic residues which releases organic acids. Organic matter, total nitrogen and available phosphorus were all higher in the topsoil than the subsoil, primarily because of the addition of organic fertilizers. While exchangeable potassium was higher in the subsoil as compared to the topsoil.

Table 6.0. Physical and chemical properties of soil in Site 2.

Parameters	Values Obtained	
	Top soil	Subsoil
Bulk density (g cm ⁻³)	1.15	1.41
Porosity (%)	56.67	46.67
pH	6.12	6.54
Organic matter (%)	3.67	0.97
Total Nitrogen (%)	0.27	0.17
Available P (ppm)	30.67	20.00
Exchangeable K (ppm)	129.00	168.67

SITE 3

General Description of Study Site 3

Site 3 was located at Sitio Ilaya Daplas, Dao, Capiz with coordinates 11° 26' 53" N latitude, 122° 32' 54" E longitude. It is surrounded by rice field, river, houses, and forest. There was no evidence of erosion in the area because it has a flat topography (Figure 3.0).



Figure 3.0. Study site 3.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 9 of 15

An interview with the owner revealed that he had been cultivating rice for almost 20 years. He also used commercial fertilizers on his rice plants. The inorganic fertilizers they applied were complete fertilizer (14-14-14), urea (46-0-0) and ammonium phosphate (16-20-0). All of the farmers applied pesticides to their rice plants. Farm activities were carried out by humans, animals, and machinery.

According to the soil map, the rice field belongs to Alimodian clay loam soil type. According to Carating et al. (2014), Alimodian clay loam is classified as Typic Hapludalfs. The soil of this series is derived from the weathered products of shale and sandstone, with shale dominating. The relief is rolling to hilly and mountainous. The external drainage is good to excessive while internal drainage is fair. The surface soil is brown to reddish brown clay loam, good medium granular structure, slightly friable when moist and brittle when dry. Round gravels and stones are present on the surface. Depth is from 20 to 30 cm. The soil is light brown clay loam, weak medium columnar structure, slightly brittle and slightly compact. Depth is from 55 to 60 cm from the surface. The substratum is gray to grayish brown, highly weathered shale sometimes weathered sandstone, weak coarse platy and slightly compact (Alicante et al. 1947). The soil profile information and description of ricefield area is presented in Table 7.0 and Table 8.0, respectively.

Table 7.0. Soil profile information of Site 3.

Profile Number	3
Soil Name	Panay river paddy
Soil Classification	Typic Hapludalf*
Location of Pedon	Sitio Ilaya, Daplas, Dao, Capiz 11° 22'42" N and 122° 41'42" E
Physiographic Position	Flat
Elevation	74.86 above sea level
Soil Moisture Regime	Udic
Soil Temperature Regime	Isohyperthermic
Present Vegetation	Rice
Farmland	Rice field
Evidence of Erosion	None
Parent Material	Sedimentary rock
Surrounding Areas	Rice field, river, houses, forest
Drainage	Good
Moisture (depths)	Moist from the surface to 20 cm; presence of aquifer at 86 cm
Stones / Rock class	Very few
Human Influence	Cultivation and land preparation, and other production practices
Date	March 20, 2022

*Adapted from Carating et al. (2014)

Table 8.0. Soil profile description of Site 3.

Horizon Designation	Depth	Horizon Description
Ap	0-25 cm	Grayish yellow brown (10YR 6/2) moist; clay; few red mottles; strong medium subangular blocky structure; firm, sticky; many very fine roots; many to fine irregular pores; common medium reddish gray concretion; slightly acidic (pH 6.24); very low nitrogen; high phosphorus; moderately unfavorable potassium; clear smooth boundary.
AB	25-98	Brown yellow (2.5 Y 7/4) moist; clay; few red mottles; moderate medium subangular blocky structure; firm, sticky; few very fine roots; few very fine irregular pores; common medium reddish gray concretion; slightly acidic (pH 6.08);

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 10 of 15

		very low nitrogen; high phosphorus; moderately unfavorable potassium; clear smooth boundary.
B	98-150 cm	Light greenish gray (7/1 N) moist; clay; many yellowish mottles; strong medium subangular blocky structure; firm, sticky; few very fine roots; common fine irregular pores; many to fine concretion; medium acidic (pH 5.68); very low nitrogen; high phosphorus; moderately unfavorable potassium; clear smooth boundary.

Soil Physical and Chemical Properties in Site 3

Presented in Table 9.0 are the physico-chemical properties of soil in Site 3. Data revealed that the bulk densities are lower which means that the soil is aggregated. Porosity values of all horizons were above 50% indicating that the soil is porous. Soil pH value is decreasing with depth manifesting that soil acidity increases as one goes down the soil profile. Soil organic matter and total N were found to be higher in the topsoil than subsoil. In terms of available phosphorus, top soil and subsoil recorded high P content based on PCAARRD (2013) criteria. For exchangeable K, the level increased with soil depth.

Table 9.0. Physical and chemical properties of soil in Site 3.

Parameters	Values Obtained	
	Top soil	Subsoil
Bulk density (g cm ⁻³)	0.86	0.78
Porosity (%)	67.55	70.57
pH	6.32	5.68
Organic matter (%)	1.78	0.54
Total Nitrogen (%)	0.09	0.03
Available P (ppm)	73.68	80.70
Exchangeable K (ppm)	3.70	20.7

SITE 4

General Description of Study Site 4

Site 4 was located at Sitio Asag, Brgy. Lonoy, Sapián, Capiz with coordinates 11.5548279° N latitude, 122.5491176° E longitude. The surrounding vegetations were rice (*Oryza sativa* L.), nipa (*Nypa fruticans*), and coconut (*Cocos nucifera*) and adjacent to a fishpond (Figure 4.0). There was no evidence of erosion in the area because it has a flat topography.



Figure 4.0. Study site 4.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 11 of 15

An interview with the farmer revealed that he had been cultivating rice for 54 years. He used synthetic fertilizers on his rice plants. The inorganic fertilizers he applied were urea (46-0-0) and ammonium phosphate (16-20-0). The farmer applied pesticide on his rice plants such as Lannete, Zennate and Maso. The average yield per cropping ranged from 10-15 sacks. Furthermore, farm activities were implemented by humans, animals and machineries.

Based on the soil map of the Philippines, the site belongs to Sopian clay soil type. According to Carating et al. (2014), Sopian soil series is classified as Typic Hapludults. The soil of this series is derived from the weathered products of granite. The surface soil is dull yellow orange when dry and gray when moist, clay, strong medium sub-angular blocky structure, firm when moist and hard when dry. The subsoil is grayish yellow when dry and gray when moist, clay loam, moderate medium sub-angular blocky structure, firm when moist and hard when dry.

The soil profile information and soil profile description of Site 4 are presented in Tables 10.0 and Table 11.0, respectively. The soil profile is shown in Figure 5.0.



Figure 5.0. Soil profile of study site 4.

Table 10.0. Soil profile information of Site 4.

Profile Number	4
Soil Name	Saline Soil
Soil Classification	Typic Hapludult
Location of the Pedon	Sitio Asag, Brgy. Lonoy, Sopian, Capiz (11.5548279° N, 122.5491176° E)
Physiographic Position	Toeslope
Elevation	67 meters above sea level
Soil Moisture Regime	Udic
Soil Temperature Regime	Hyperthermic
Present Vegetation	Rice (<i>Oryza sativa</i> L.)
Farmland	Ricefield
Evidence of Erosion	None

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 12 of 15

Parent Material	Granite
Surrounding Areas	Rice farms, fish pond, nipa plantation, coconut plantation
Drainage	Moderately drained
Moisture (Depths)	Moist from the surface up to 41cm, presence of aquifer at 70 cm
Stones/Rocks Class	No stone on the surface, rocky on the substratum
Human Influence	Cultivation, farming
Date	March 17, 2024

Table 11.0. Soil profile description of Site 4.

Horizon Designation	Depths	Horizon Description
A	0-41 cm	Dull yellow orange (10YR 7/3) dry, gray (2.5 YR 5/6) moist; clay; common red, orange and gray mottles; strong, medium sub-angular blocky structure; firm when moist and very hard when dry, slightly sticky; non-plastic; many fine roots; few micropores to fine irregular pores; near neutral (pH 6.76); very low nitrogen; very low phosphorus; medium potassium; gradual irregular boundary.
B1	41-68 cm	Grayish yellow brown (10 YR 5/2) dry, gray (2.5 YR 3/2) moist; clay loam; common red, orange, and gray mottles; strong medium sub-angular blocky structure, firm when moist and hard when dry, non-sticky; non-plastic; few very fine roots; few micropores to coarse tubular pores; slightly acidic (pH 6.58); low nitrogen; medium phosphorus; low potassium; gradual irregular boundary.
B2	68-138 cm	Dark (5Y 4/3) dry, olive black (5Y 3/2) moist; silty clay loam; common grayish mottles; moderate, medium sub-angular blocky structure, soft when dry and firm when moist, non-sticky; non-plastic; few fine tubular pores; extremely acidic (pH 3.46); very low nitrogen; very low phosphorus; medium potassium; gradual irregular boundary.
BC	Below 138 cm	Brownish gray (7.5 YR 5/1) dry, greenish black (2.5 GY 2/1) moist; loam; moderate medium sub-angular blocky structure, soft when dry, firm when moist, slightly sticky; non-plastic; extremely acidic (pH 3.29); medium nitrogen; low phosphorus; very low potassium.

Soil Chemical Properties

Table 12.0 presents the chemical properties of soil in Site 4. In terms of soil pH, the results revealed that the top soil had a pH of 6.76 which is classified as near neutral while the subsoil had a lower pH of 5.02 which is classified as strongly acid, indicating more acidity in the deeper layers. These results conform to the description of Sparks (1995) that since electrolyte concentration is high, the soil pH of saline soil is usually <8.5 and the soil is flocculated. Likewise, mottles were observed in the horizons as irregular spots of colors in soil that can indicate variations in soil pH levels (FAO, 2017). The presence of red mottles in the top soil has led to a near-neutral pH, while the subsoil contains more grey mottles resulting in a lower pH level.

In terms of soil OM, results revealed that top soil had a 2.78% OM which is classified as very low while the subsoil had a higher percentage of OM equivalent to 7.36%, which is classified as medium (PCAARRD, 2013). This findings indicate better OM content in the deeper layers.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 13 of 15

The top soil had 0.08% total N which is classified as very low, while the subsoil showed a higher concentration of total N amounting to 0.10% classified as low (PCAARRD, 2013). This indicates more N content in the subsoil than the top soil.

Moreover, the top soil had 8 ppm of available phosphorus (P) which is classified as very low while the subsoil showed a much higher value of 28.33 ppm available P which is classified as medium (PCCAARD, 2013). This indicates more P availability at the subsurface of the soil profile.

Table 12.0. Chemical properties of soil in Site 4.

Parameters	Values Obtained	
	Top soil	Subsoil
pH	6.76	5.02
Organic matter (%)	2.78	7.36
Total Nitrogen (%)	0.08	0.10
Available P (ppm)	8.00	28.33
Exchangeable K (ppm)	0.06	0.05
Electrical conductivity (µS/cm)	18,126.67	

Both top soil and subsoil had very low levels of exchangeable potassium (K), with values of 0.06 ppm and 0.05 ppm, respectively. This low concentration of K⁺ may be due to the dominance of Na⁺ in the exchange site of saline soil.

In terms of electrical conductivity (EC), the water sample collected during soil profiling recorded an EC value of 18,126.67 µS/cm or 18 dS/m. According to Sparks (1995), saline soils are those with EC from the saturation extract amounting to >4 dS/m. This implies that the site qualifies the criteria of a saline soil.

Likewise, another unique observation during profile characterization is the presence of a highly acidic subsoil from 68 cm downwards. This extremely acidic subsurface layer could be considered a potential acid sulfate soil. According to Attanandana & Vacharotayan (1986), acid sulfate soils are soils with a pH below 4 that is directly or indirectly caused by sulfuric acid formed by oxidation of pyrite (FeS). The abundance of sulfur could be found in the very high organic matter content which is yet to be mineralized. Additionally, majority of acid sulfate soils occur in coastal areas developed from recent or semi-recent sediments. They are usually restricted to areas relatively close to the sea.

Soil Quality Improvement Program

The data obtained indicate the existence of variabilities and commonalities among identified paddy soils in the province of Capiz. Thus, the following site-specific management strategies to improve the quality of soil for crop production are formulated.

Table 13.0. Recommended management strategies for each paddy site.

Paddy Site	Management Strategies
No. 1	• Reduce tillage farming method • Provision of clean irrigation water • Ensuring adequate drainage • Proper rice straw management • Incorporation of liming materials • Application of fertilizers
No. 2	• Reduce tillage farming method • Provision of clean irrigation water • Ensuring adequate drainage

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 14 of 15

	• Monitor soil pH and apply liming materials once needed • Application of N fertilizers • Adding organic matter to the soil
No. 3	• Apply organic residues and farm manures to restore fertility of the soil • Apply adequate N and K fertilizers • Practice minimum tillage • Proper timing of planting to avoid destruction of crops suing seasonal flooding
No. 4	• Planting salt tolerant rice varieties • Provision of clean irrigation water • Ensuring adequate drainage • Reduce tillage farming method • Application of fertilizer

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate hedings if they are long. Justify paragraphs with main headings in caps.)

Based from the findings of this study, the following conclusions are drawn:

- The different paddy soils under study varied in terms of their horization and morphological features.
- Paddy soils which were characterized in this study differed in their physical and chemical properties attributed to the type of cultural and management practices employed by the tenant farming the land.

The following recommendations are forwarded:

- Employ the package of management recommendations outlined in Table 13.0 to address the limitations and properties identified in each site and to make the soil more favorable for rice production.
- Conduct monitoring studies to track the changes in the morphological, physical, chemical and biological properties of soil determined in each site.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Package of management recommendations

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

The author would like to thank the CAPSU CSRDC for allocating budget for the procurement of some supplies and materials used in this research.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 15 of 15

Attanandana, T., & Vacharotayan, S. (1986). Acid Sulfate Soils: Their Characteristics, Genesis, Amelioration and Utilization. Japanese Journal of Southeast Asian Studies, 24(2), 154-180

Carating, R. B., Galanta, R. G., & Bacatio, C. D. (2014). The soils of the Philippines. Springer Science & Business

Food and Agriculture Organization (FAO) (2017). Soil Fertility and Plant Nutrition in the Philippines. Retrieved from <https://www.fao.org/3/a-i6902e.pdf>

PCAARRD. (2013). The Philippines Recommends for Soil Diagnosis and Amelioration. 138p. Los Baños, Laguna: PCARRD- (Philippines Recommends Series No.93/2007)

Sparks, D. L. (1995). The chemistry of saline and sodic soils. Environmental soil chemistry, 219-232.

United States Department of Agriculture - Natural Resources Conservation Service. (n.d.). Retrieved from <https://www.nrcs.usda.gov/>